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(54) **MANAGING TRAFFIC RULES IN ASSOCIATION WITH FULLY QUALIFIED DOMAIN NAMES (FQDNS)**

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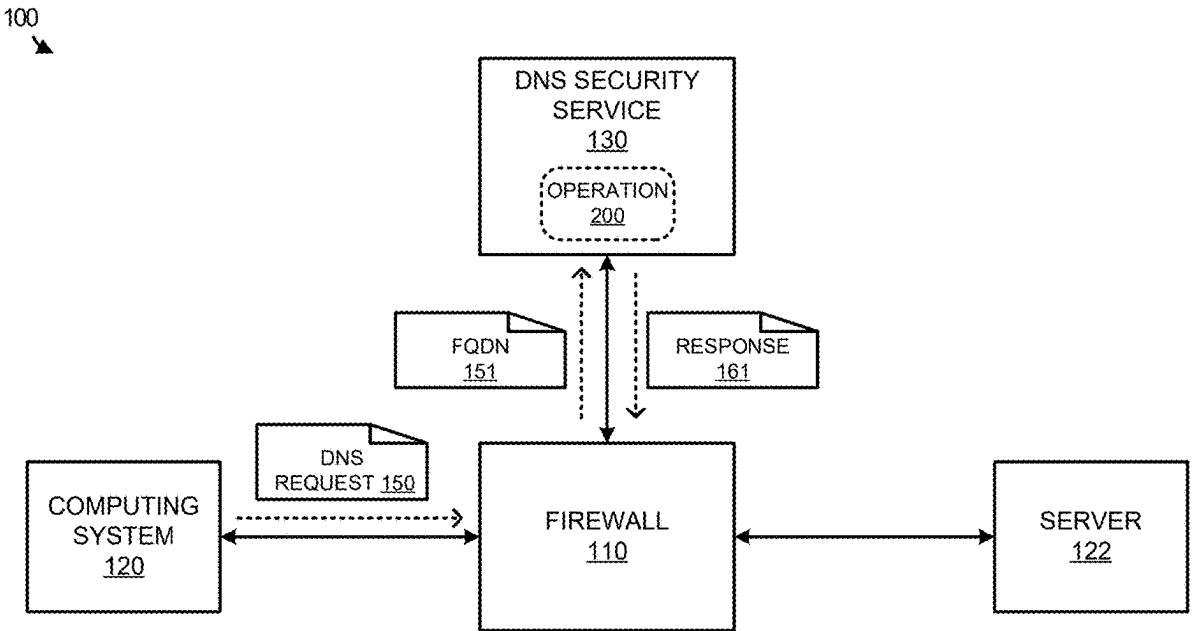
(60) Provisional application No. 63/211,448, filed on Jun. 16, 2021.

Publication Classification

(51) **Int. Cl.**
H04L 9/40 (2022.01)

(57) **ABSTRACT**

Provided herein are systems, methods, and software to manage traffic rules in association with fully qualified domain names (FQDNs). In one example, a method includes receiving a FQDN identified in a domain name system (DNS) request, determining trust factors indicative of reliability and reputation of the FQDN, and updating a trust score for the FQDN in accordance with the trust factors. The trust score was generated from previously determined trust factors for the FQDN in response to prior DNS requests received before the DNS request. After updating the trust score, the method includes determining the trust score satisfies an action threshold of a plurality of action thresholds and implementing a traffic rule, of a plurality of traffic rules, corresponding to the action threshold. Different traffic rules of the plurality of traffic rules correspond to different action thresholds of the plurality of action thresholds.



100

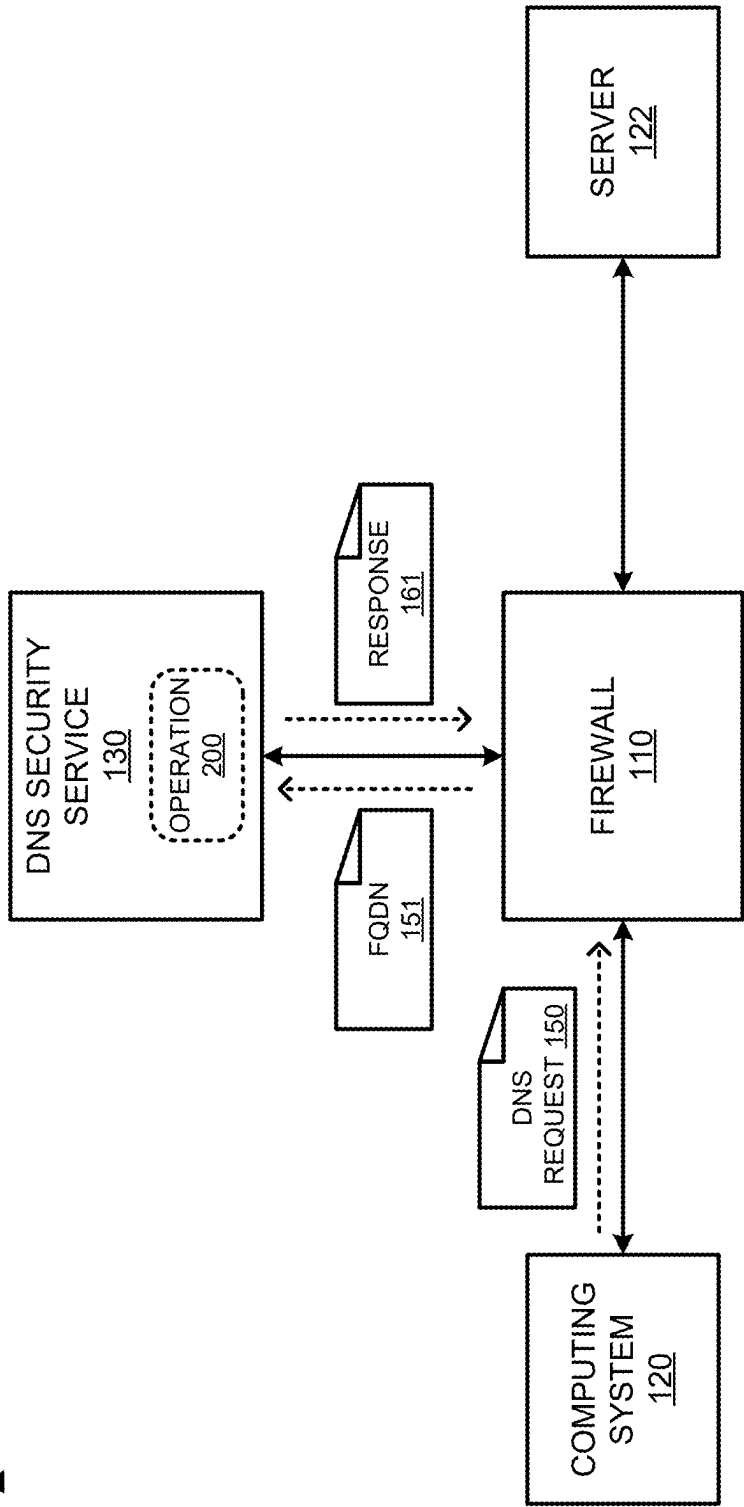


FIGURE 1

200

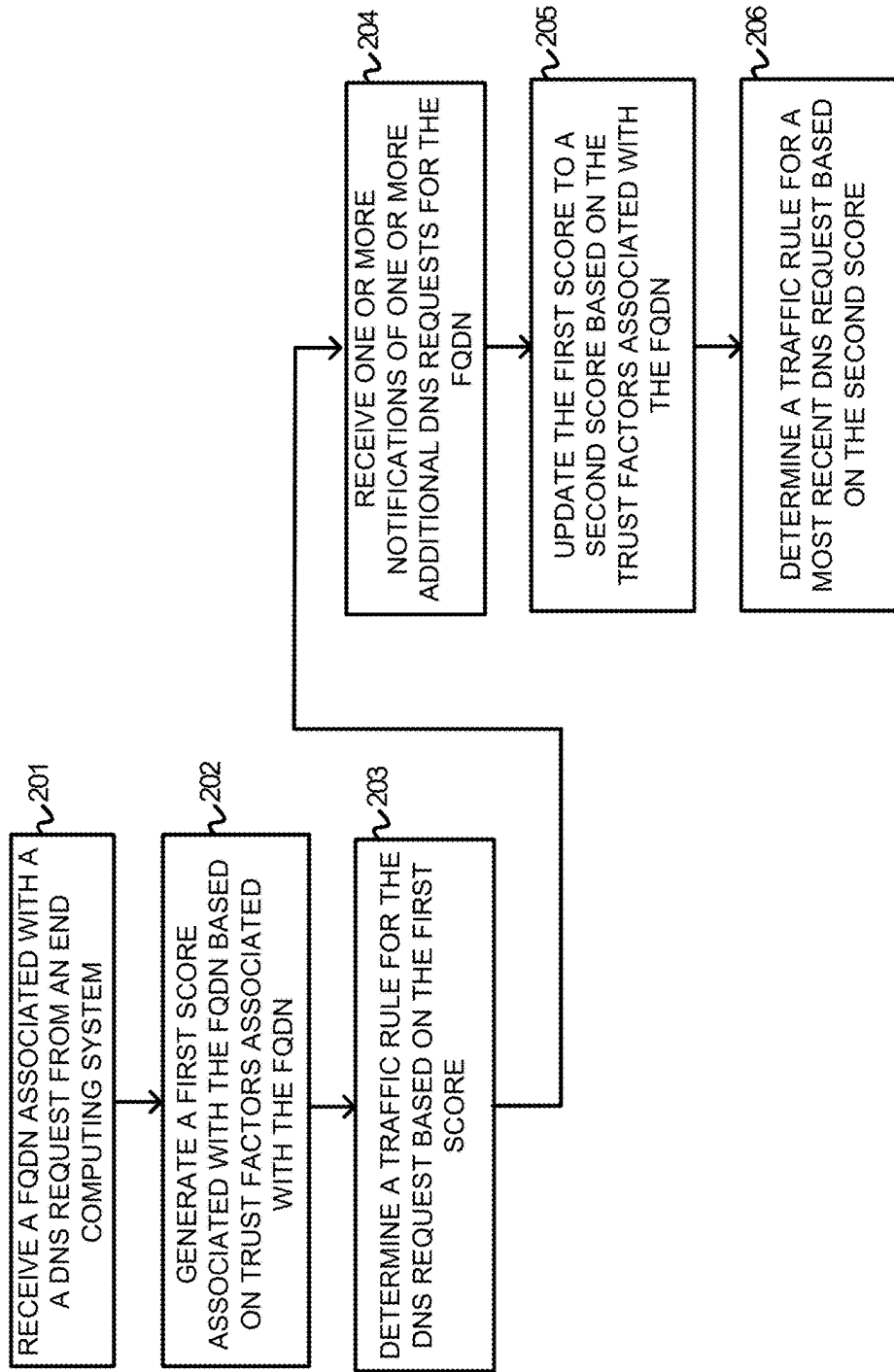


FIGURE 2

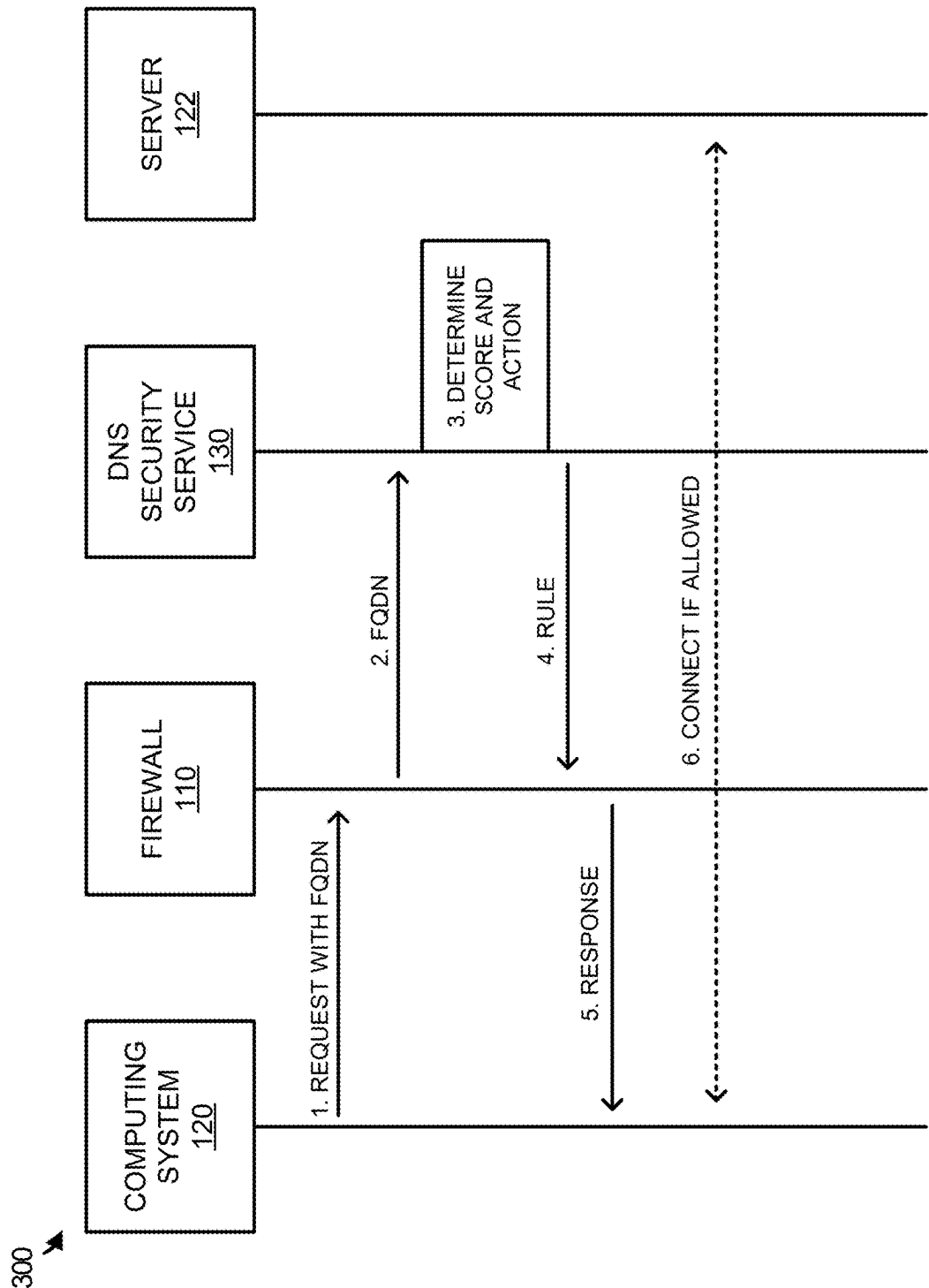


FIGURE 3A

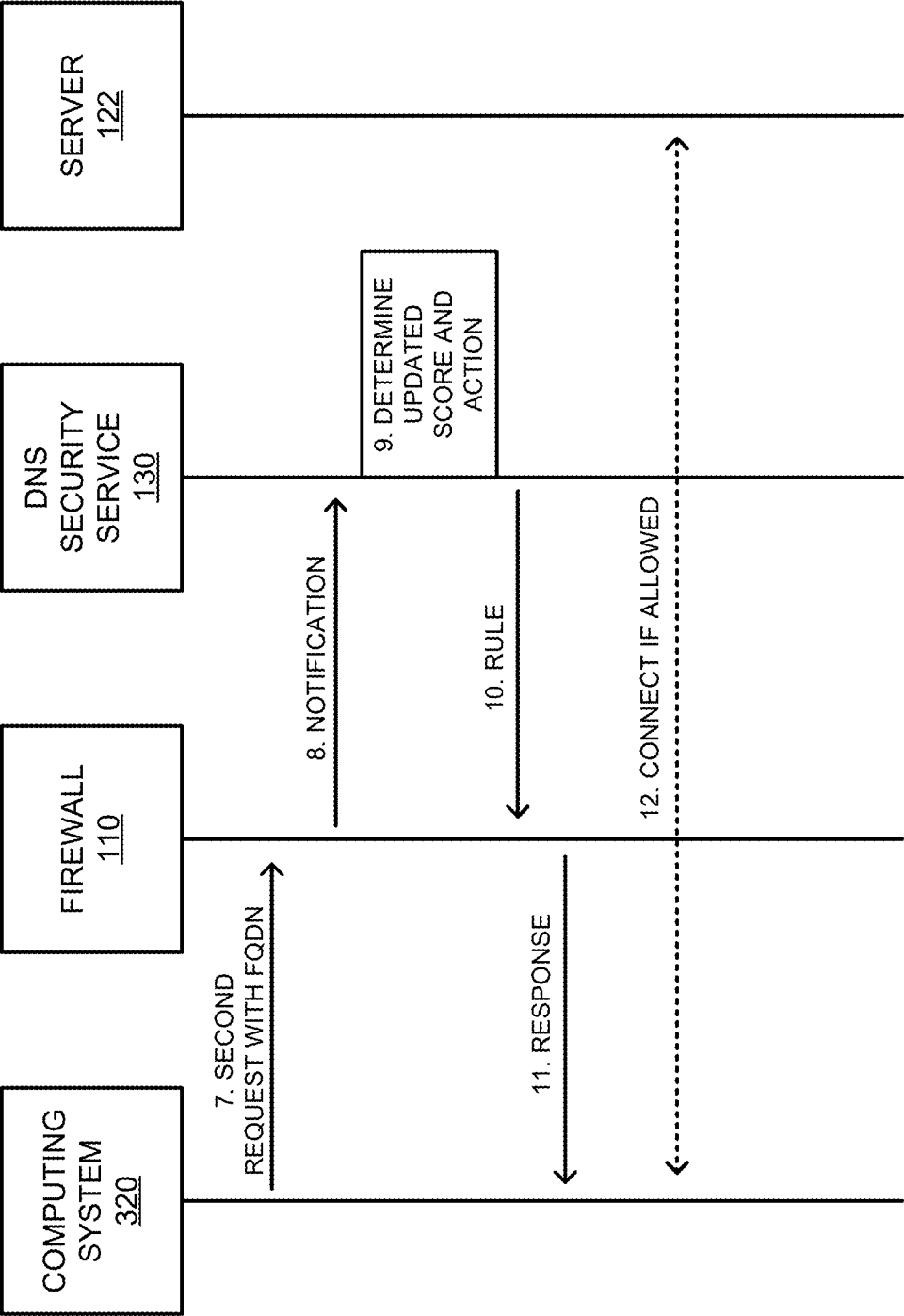


FIGURE 3B

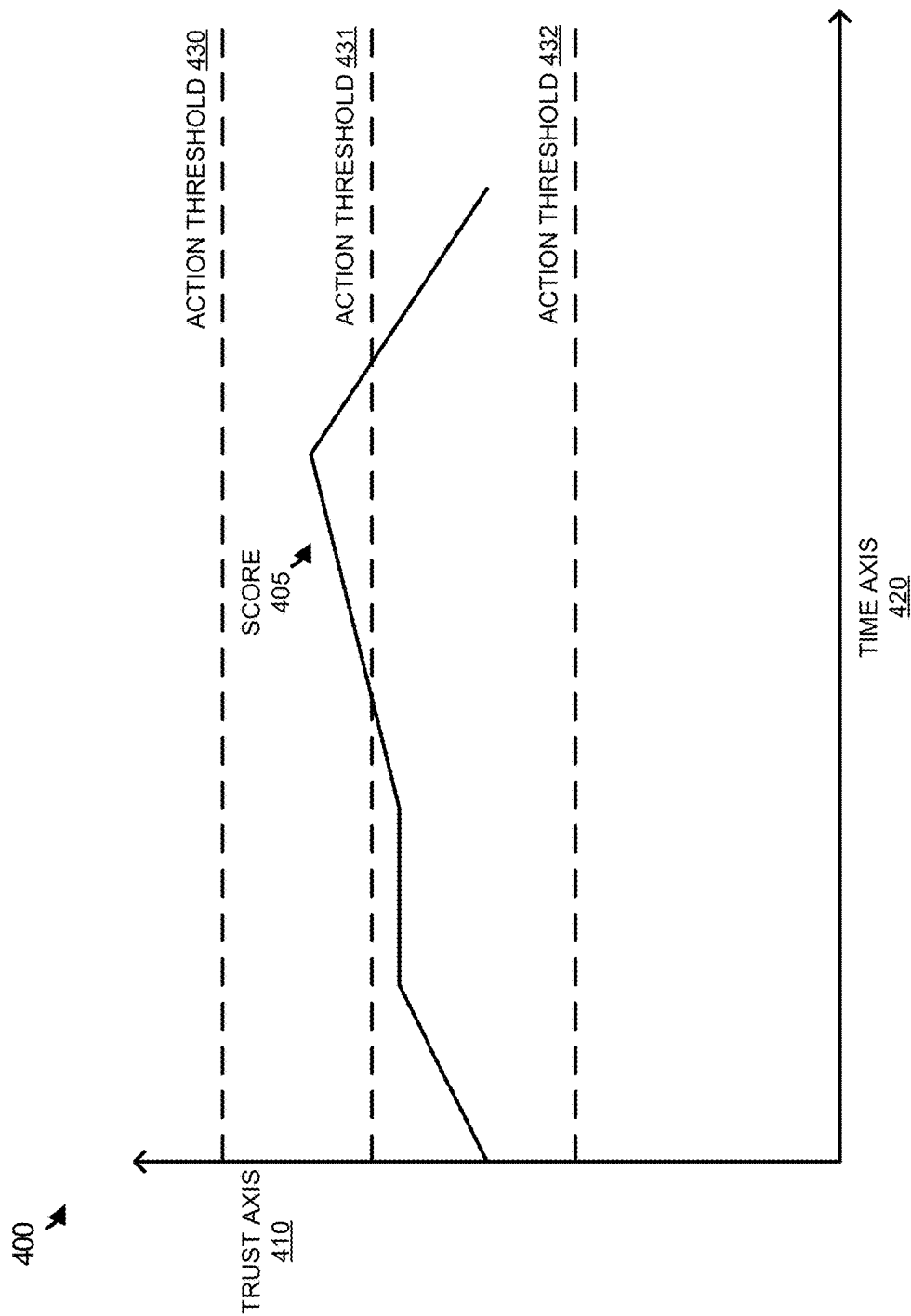


FIGURE 4

500

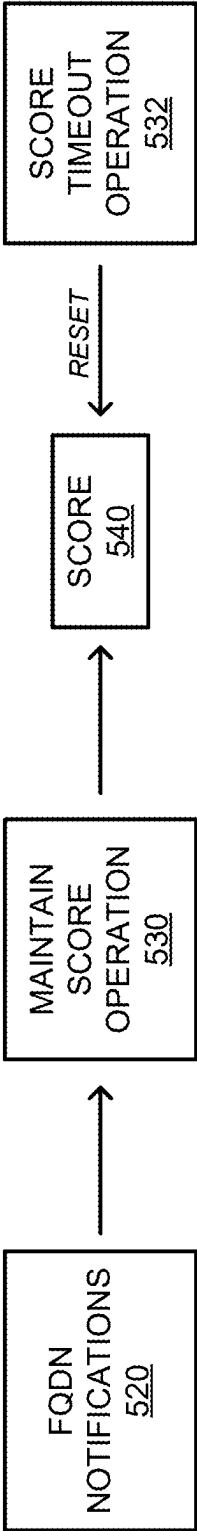


FIGURE 5

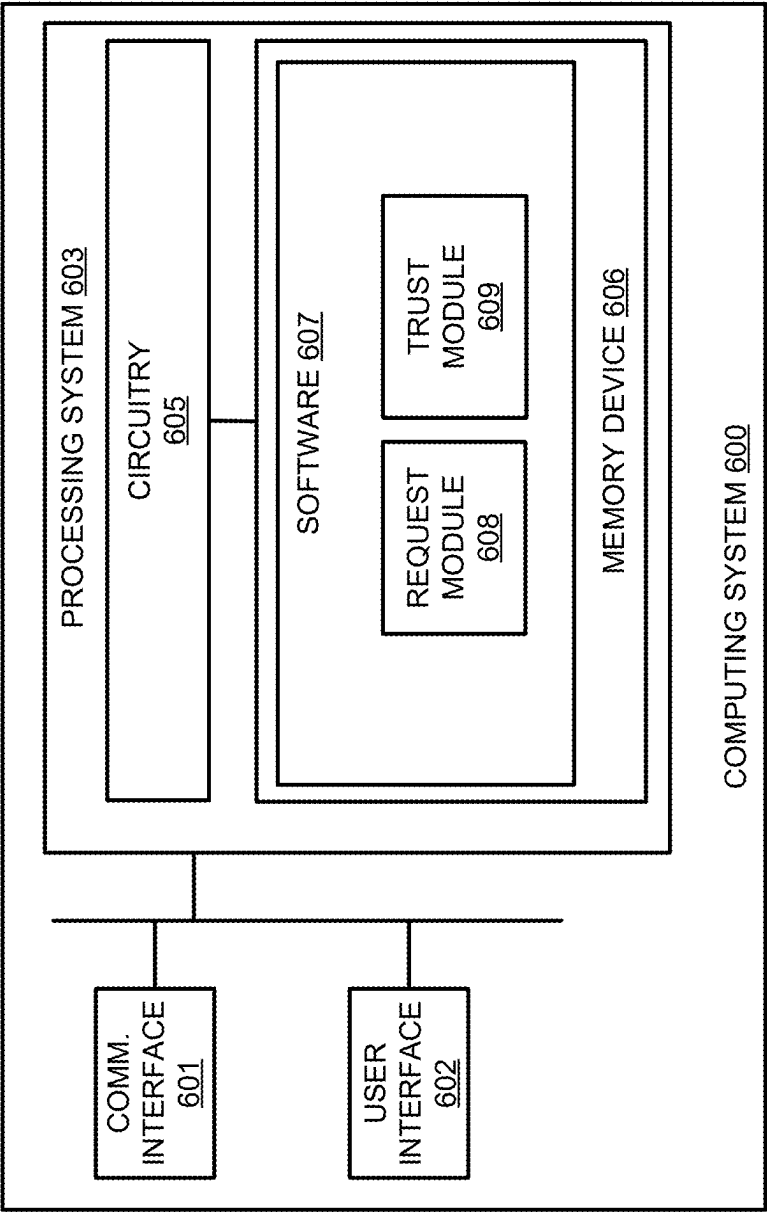


FIGURE 6

MANAGING TRAFFIC RULES IN ASSOCIATION WITH FULLY QUALIFIED DOMAIN NAMES (FQDNS)

RELATED APPLICATIONS

[0001] This application is a continuation of and claims priority to U.S. Pat. No. 12,309,118, entitled “MANAGING TRAFFIC RULES IN ASSOCIATION WITH FULLY QUALIFIED DOMAIN NAMES (FQDNS),” filed on Jun. 15, 2022, which is related to and claims priority to U.S. Provisional Patent Application No. 63/211,448, entitled “MANAGING TRAFFIC RULES IN ASSOCIATION WITH FULLY QUALIFIED DOMAIN NAMES (FQDNS),” filed on Jun. 16, 2021, and which are both hereby incorporated by reference in their entirety.

BACKGROUND

[0002] In computing environments, firewall services may be used to monitor incoming and outgoing network traffic to permit or block data packets based on security rules. In some examples, some firewall services may provide a Domain Name Server (DNS) resolver that is used to receive DNS requests from computing systems on a local network and provide the requesting computing system with a corresponding internet protocol (IP) address.

[0003] In some examples, DNS requests from computing systems may include fully qualified domain names (FQDNs). A FQDN specifies the exact location in the tree hierarchy of the DNS. It specifies all domain levels, including the top-level domain and the root zone. In some examples, the FQDN can have a host name and at least one higher-level domain (label) and always ends in the top-level domain. When a request is received from a computing device, a resolver may identify an IP address for the connection and provide the IP address to the requesting device. However, difficulties can arise in determining whether the connection with the IP address for the FQDN is malicious and whether a connection should be permitted with the destination associated with the FQDN.

Overview

[0004] Provided herein are systems, methods, and software to manage traffic rules in association with fully qualified domain names (FQDNs). In one example, a method includes receiving a FQDN identified in a domain name system (DNS) request, determining trust factors indicative of reliability and reputation of the FQDN, and updating a trust score for the FQDN in accordance with the trust factors. The trust score was generated from previously determined trust factors for the FQDN in response to prior DNS requests received before the DNS request. After updating the trust score, the method includes determining the trust score satisfies an action threshold of a plurality of action thresholds and implementing a traffic rule, of a plurality of traffic rules, corresponding to the action threshold. Different traffic rules of the plurality of traffic rules correspond to different action thresholds of the plurality of action thresholds.

[0005] In other examples, at least one apparatus is provided to perform steps like those in the method above.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] Many aspects of the disclosure can be better understood with reference to the following drawings. While several implementations are described in connection with these drawings, the disclosure is not limited to the implementations disclosed herein. On the contrary, the intent is to cover all alternatives, modifications, and equivalents.

[0007] FIG. 1 illustrates a computing environment to manage permissions associated with fully qualified domain names (FQDNs) according to an implementation.

[0008] FIG. 2 illustrates an operation of a domain name system (DNS) security service to manage permissions associated with FQDNs according to an implementation.

[0009] FIG. 3A-3B illustrate a timing diagram of managing permissions associated a FQDN according to an implementation.

[0010] FIG. 4 illustrates a graph demonstrating the trust score for a FQDN according to an implementation.

[0011] FIG. 5 illustrates an operational scenario of updating a trust score associated with a FQDN according to an implementation.

[0012] FIG. 6 illustrates a DNS security service computing system according to an implementation.

DETAILED DESCRIPTION

[0013] In many computing environments, firewalls are deployed that are used to monitor ingress and egress packets for one or more computing systems and determine whether to block, permit, redirect, or provide some other operation with respect to the packet. These determinations may be based on rules that can act on packets based on addressing in the packet, protocols for the packet, or some other factor. Here, a firewall may identify domain name system (DNS) requests from a computing system or device in the network that includes a fully qualified domain name (FQDN). A FQDN specifies the exact location in the tree hierarchy of the DNS. It specifies all domain levels, including the top-level domain and the root zone. In some examples, the FQDN can have a host name and at least one higher-level domain (label) and always ends in the top-level domain. Once the FQDN is identified, the FQDN may be provided to a DNS security service that analyzes infrastructure associated with the FQDN to determine a traffic rule for FQDN. The traffic rule may be used to permit, block, redirect, or provide some other action in association with the destination server for the FQDN.

[0014] To analyze the infrastructure associated with the FQDN, the DNS security service may analyze trust factors associated with the reputation of the DNS response for the FQDN. These trust factors may be associated with the A or AAAA records for the FQDN, the authoritative name server associated with the FQDN, the number of requests with the FQDN, or some other factor. The information for the trust factors may be maintained locally but may also be obtained from other services or databases that can provide information on the status, reputation, location, or some other information associated with the trust factors. Advantageously, rather than identifying information associated with the destination server or destination IP address, the DNS security service may process information about the reputation of the DNS response, including the services, certificates, records, and systems providing the response, to determine if the requested destination is potentially malicious. In some

examples, the factors may be used to generate a trust score for the FQDN. Each of the factors may contribute to the trust score uniformly or may contribute differently to the trust score. For example, a risk assessment for a Name Server may contribute differently to the trust score than the number of requests for the FQDN.

[0015] Once the score is determined, a traffic rule may be identified from a set of traffic rules based on the score satisfying criteria for that rule. The traffic rule may then be provided to the firewall to implement the action corresponding to the rule. In implementing the traffic rule, the firewall may prevent a DNS response from being provided for the DNS request, may permit the DNS response to be provided to the requesting computing system, may block traffic associated with the IP address retrieved using the DNS request, may redirect traffic to another IP address, or may perform some other action.

[0016] FIG. 1 illustrates a computing environment **100** to manage permissions associated with FQDN according to an implementation. Computing environment **100** includes computing system **120**, firewall **110**, domain name system (DNS) security service **130**, and server **122**. Computing environment **100** includes DNS request **150**, fully qualified domain name (FQDN) **151**, and response **161**. DNS security service **130** is configured to provide at least operation **200** that is described below with respect to FIG. 2.

[0017] In operation, firewall **110** may be used to provide security associated with computing systems or devices that attempt to connect over the network. The firewall monitors ingress and egress packets for the computing system and determines whether to allow, block, redirect, or provide some other operation in association with a packet or communication. Here, computing system **120** generates a DNS request **150** that includes a FQDN. A FQDN specifies the exact location in the tree hierarchy of the DNS. It specifies all domain levels, including the top-level domain and the root zone. In some examples, the FQDN can have a host name and at least one higher-level domain (label) and always ends in the top-level domain.

[0018] When the request is received by firewall **110**, firewall **110** may provide or communicate the FQDN to DNS security service **130**. DNS security service **130** is used to determine whether the FQDN requested from computing system **120** should be trusted or that the destination is likely malicious. To make this determination, DNS security service may calculate a trust score that can be used to select a traffic rule or action for the request, wherein the action may include permitting the DNS request and the communication, blocking the request and the communication, redirecting the request or communication, or providing some other action in association with the request. In generating the score, the DNS security service **130** may consider a variety of trust factors associated with the FQDN. The trust factors may include time delta information between requests for the FQDN, a volume of queries for the FQDN, infrastructure information associated with the A or AAAA records associated with the FQDN, a risk score of the authoritative Name Server, a quantity of record changes associated with the FQDN, SSL certificate comparison with other domains, a quantity of TXT records associated with the FQDN, or some other factor. The infrastructure information for the A or AAAA records may include information about the time to live associated with the FQDN, location of the record, or some other information associated with the record. The risk

score of the authoritative Name Server may be determined based on reputation information provided from one or more sources and may be determined based on length of operation, owner, or some other information.

[0019] In some implementations, each of the factors may contribute to the score, wherein the score may comprise a numerical value, a letter grade, or some other scoring mechanism. For example, for a first quantity of requests for the FQDN, DNS security service **130** may allocate a first score, while for a second quantity of requests for the FQDN, DNS security service **130** may allocate a second score. The scores may then be aggregated for the various factors to determine a current score for the FQDN. Once the score is calculated in association with the FQDN, DNS security service **130** may determine traffic rules for the DNS request based on the score. In some implementations, different traffic rules may be deployed based on whether the score satisfies criteria or a threshold. For example, if a score satisfies a first score threshold, DNS security service **130** may permit the connection using the address associated with the FQDN. However, if the score satisfies a second score threshold, DNS security service **130** may block the connection with an associated IP address.

[0020] FIG. 2 illustrates an operation **200** of a domain name system (DNS) security service to manage permissions associated with FQDNs according to an implementation. The steps of operation **200** are referenced parenthetically in the paragraphs that follow with reference to systems and elements of computing environment **100** of FIG. 1.

[0021] As depicted, operation **200** includes obtaining **(201)** a FQDN associated with a DNS request by a computing system. The DNS request may comprise a request from a physical computing device, such as a desktop computer, laptop computer, and the like, or may comprise a request from a virtual computing device, such as a virtual machine or container. When an FQDN is identified as part of the request by a firewall **110**, the FQDN may be forwarded to DNS security service **130**. Although demonstrated as separate in the example of FIG. 1, the operations of DNS security service **130** may be implemented wholly or partially at the computing system providing the firewall.

[0022] In response to obtaining the FQDN, operation **200** further generates **(202)** a first score for the FQDN based on trust factors associated with the FQDN. The trust factors may include, but are not limited to, time delta information between requests for the FQDN, a volume of queries for the FQDN, infrastructure information associated with the A or AAAA records associated with the FQDN, a risk score of the authoritative Name Server, a quantity of record changes associated with the FQDN, SSL certificate comparison with other domains, a quantity of TXT records associated with the FQDN, or some other factor. For each of the factors, a score may be determined that can contribute to the overall score associated with the FQDN. For example, a first authoritative Name Server may be allocated a first score based on its reputation (e.g., length of operation, owner, etc.), while a second authoritative Name Server may be allocated a second score based on its reputation. The score for the individual factor may be combined with scores associated with one or more other factors to determine the overall score associated with the FQDN. In some implementations, scores for some factors may weigh more heavily than scores associated with other factors. For example, the risk score associated with the authoritative Name Server may contribute more to the

overall score for the FQDN than the score associated with the volume of queries for the FQDN.

[0023] Once the score is determined for the FQDN, DNS security service **130** may determine **(203)** a traffic rule from a plurality of traffic rules for the DNS request based on the first score. The traffic rule may permit the connection to the address associated with the FQDN, block the connection to the address associated with the FQDN, redirect the connection to another destination, or provide some other action in association with the FQDN. Once determined, DNS security service **130** may provide the traffic rule to the firewall to apply the rule. In some examples, the firewall may return the requested address, block the return of the DNS to the requesting computing device, may provide a redirect IP address to the requesting computing device, or may implement some other action to manage the connection for the computing device.

[0024] After generating the first score, operation **200** further obtains **(204)** one or more notifications of one or more additional DNS requests for the FQDN. In response to the one or more notifications, operation **200** updates **(205)** the first score to a second score based on the trust factors associated with the FQDN. In some implementations, the trust score is updated after each of the DNS requests with the FQDN, wherein the score may increase or decrease based on changes in association with the various factors.

[0025] Once the second score is determined, operation **200** determines **(206)** a second traffic rule from the plurality of traffic rules for a most recent DNS request of the one or more additional DNS requests. The score may be updated based on the changes associated with the various factors. For example, a FQDN may initially use a first authoritative Name Server that is associated with a first score, while, during a second DNS request, the FQDN may use a second authoritative Name server that is associated with a second score. The changes in the score for the Name Server may be used to update the overall score that is used to determine the traffic rule for the FQDN. In some implementations, each of the trust factors may be updated for each DNS request with the FQDN. In other implementations, a portion of the trust factors may be updated or checked for each DNS request with the FQDN. As an example, the authoritative Name Server may only be checked once for every five DNS requests with the FQDN, while the statistics for the number of requests with the FQDN may be updated for each of the requests.

[0026] In some examples, as the score is updated based on for the FQDN, DNS security service **130** may determine when the traffic rule changes for the FQDN. DNS security service **130** may indicate the current traffic rule after each DNS request, may update firewall **110** on the traffic rule only when the traffic rule changes, or may provide information about the traffic rule on some other interval.

[0027] In some implementations, in determining the traffic rule, DNS security service **130** may determine when the score satisfies one or more criteria, or a threshold associated with a traffic rule. For example, a first score or a low trust score may qualify for a traffic rule to block communications associated with the FQDN, while a second score or high trust score may qualify for a traffic rule to permit connections associated with the FQDN.

[0028] In some examples, the score associated with the FQDN may expire or time out, resulting in the score being reset. The expiration of the FQDN may occur in response to

a period expiring after the first DNS request with the FQDN, may occur in response to a period expiring after the most recent DNS request with the FQDN, or may occur at some other interval. For example, if a period expires following the most recent DNS request with the FQDN, DNS security service **130** may reset the score associated with the FQDN. When a new request is obtained, FQDN may calculate a new score using new information, which may include resetting counters, request statistics, or some other information associated with the FQDN.

[0029] FIGS. **3A-3B** illustrate timing diagram **300** of managing permissions associated a FQDN according to an implementation. The timing diagram in FIGS. **3A** and **3B** include computing system **120**, firewall **110**, DNS security service **130**, and server **122** from computing environment **100** of FIG. **1**. FIG. **3B** includes computing system **320**, which is representative of a computing system that uses firewall **110** for managing connection limitations.

[0030] Referring first to FIG. **3A**, computing system **120**, generates a DNS request with a FQDN, at step **1**, that is received by firewall **110**. In response to the request, firewall **110** may provide, at step **2**, the FQDN to DNS security service **130** for a trust analysis of the FQDN. DNS security service **130** then determines, at step **3**, a score, and action (traffic rule) for the FQDN based on trust factors associated with the DNS target of the FQDN. The trust factors are associated with the service or infrastructure that is providing the DNS lookup associated with the FQDN. The trust factors may include time delta information between requests for the FQDN, a volume of queries for the FQDN, infrastructure information associated with the A or AAAA records associated with the FQDN, a risk score of the authoritative Name Server, a quantity of record changes associated with the FQDN, SSL certificate comparison with other domains, a quantity of TXT records associated with the FQDN, or some other factor. Additional factors may include the number of requests for the FQDN from other networks not supported by firewall **110**, databases indicating potentially malicious Name Servers, or some other source of information related to the trust of the FQDN. The factors may be used to identify risks in association with the publishing or records of the FQDN as part of DNS, which may indicate that the end IP address is malicious. In some examples, each of the factors may include a contributory score to the overall score for the FQDN. Each of the factors may be weighted evenly or may be weighted differently in contributing to the overall score. For example, a contributory score related to the authoritative name server may contribute more to the overall score than the quantity of requests for the FQDN.

[0031] Once the score is determined for the FQDN, an action or traffic rule for the request is identified and provided, at step **4**, to firewall **110**. The traffic rule may be identified based on the score satisfying one or more criteria associated with the traffic rule. The traffic rule may indicate to firewall **110** whether to block, permit, redirect, or perform some other action in association with the DNS request. In some implementations, if the DNS request is blocked in association with the FQDN, then firewall **110** may never provide an IP address to the requesting computing system. In other examples, where the request is permitted for the FQDN, firewall **110** may provide the IP address identified during a DNS lookup. In still further examples, if the request for the FQDN should be redirected, firewall **110** may provide a different IP address in place of the IP address for the

FQDN. This alternative IP address may be used to indicate to the requesting computing system that the requested destination server could be malicious. Here, firewall **110** provides a response, at step **5**, to computing system **120** for the DNS request and computing system **120** initiates, at step **6**, a connection with server **122** if the connection is allowed.

[0032] In some implementations, rather than blocking the DNS response to computing system **120** based on a traffic rule, firewall **110** may provide the IP address for the DNS request, but block, terminate, or reset the connection with the server to prevent communications between computing system **120** and server **122**. In some examples, firewall **110** may cache the most recent traffic rule identified for a FQDN and enforce the rule until an update is provided by DNS security service **130**.

[0033] Turning to FIG. 3B, computing system **320** generate and communicate second DNS requests with the FQDN to firewall **110** at step **7**. In turn, firewall **110** provides notifications to DNS security service **130** that the DNS requests were received with the FQDN at step **8**. In response to receiving the notifications, DNS security service **130** may update the score associated with the FQDN and update the traffic rule or action associated with the FQDN at step **9**. In some implementations, DNS security service **130** may update each of the factors for the FQDN for each DNS request from computing system(s) **320**. In other implementations, a portion or subset of the factors may be updated for the FQDN for each request. For example, the factor related to the quantity of requests for the FQDN might be updated during each request, while the factor associated with the authoritative Name Server may be updated after five requests. Once the updates are identified for the individual factors, the overall score may be updated and used to identify a traffic rule for the FQDN. In some examples, the second traffic rule identified for the DNS request from computing system **320** may comprise a different traffic rule than the request

[0034] Once the traffic rule is identified, the traffic rule is communicated, at step **10**, to firewall **110** for enforcement. In the example of a rule that permits traffic associated with the FQDN, firewall **110** may provide a DNS response that includes the IP address associated with the FQDN and computing system **320** may initiate a connection with server **122**, at step **12**, when the connection is allowed.

[0035] In other examples, if the FQDN is identified to be potentially malicious, firewall **110** may not provide a response to the DNS request, may indicate an IP address associated with a redirect instead of the desired destination, or may provide some other response (or lack of response) based on the rule provided by DNS security service **130**. In some implementations, firewall **110** may cache the rule for the FQDN and may apply the rule until an update is received from DNS security service **130**. In some examples, firewall **110** may enforce rules on the IP addresses returned from the DNS request to block or allow communications based on the traffic rule provided from DNS security service **130**. For example, after a FQDN is provided to DNS security service **130**, DNS security service **130** may determine that the server is potentially malicious and provide an indication to firewall **110**. As a result of the notification, firewall **110** may block, reset, or terminate connections associated with the IP address for the DNS lookup.

[0036] FIG. 4 illustrates a graph **400** demonstrating the trust score for a FQDN according to an implementation.

Graph **400** includes trust axis **410**, time axis **420**, action thresholds **430-432**, and score **405**.

[0037] As described herein, a DNS security service may be used to obtain FQDNs from at least one firewall and determine a trust score associated with the FQDN. The trust score is generated from factors associated with the DNS target and DNS resolution information associated with the FQDN. The factors may include time delta information between requests for the FQDN, a volume of queries for the FQDN, infrastructure information associated with the A or AAAA records associated with the FQDN, a risk score of the authoritative Name Server, a quantity of record changes associated with the FQDN, SSL certificate comparison with other domains, a quantity of TXT records associated with the FQDN, or some other factor. The factors may be used to estimate the trust of the destination server by using infrastructure information associated with how a DNS request with the FQDN is managed (Name Server information, record information, etc.). In some implementations, each of the factors may contribute equally to the overall score for the FQDN, however, the factors may contribute unequally to the score for the FQDN.

[0038] After calculating a first score for the FQDN, the DNS monitoring system may update the score based on subsequent requests that include the FQDN. The updates may be based on the number of requests with the FQDN, the frequency of the requests with the FQDN, changes in the status of the records or Name Servers, or some other updates. In some implementations, each of the factors are updated after each request, but one or more of the factors may be updated periodically, after a quantity of requests, or at some other interval. When the factors are updated, the new score is calculated and used to determine the traffic rule for the FQDN.

[0039] Here, different traffic rules are triggered via action thresholds **430-432**, wherein the thresholds comprise values associated with the trust score. For example, the score **405** may be calculated on a scale of 0-100, action threshold **432** may comprise a value of 50, action threshold **431** may comprise a value of 80, and action threshold **430** may comprise a value of 90. When the score satisfies a threshold, the traffic rule associated with that threshold may be triggered. The traffic rule may then be provided to the firewall to implement the corresponding action, which may include preventing a DNS response from being provided to the device, permitting a DNS response to be provided to the device, blocking or termination connections with the IP address retrieved for the FQDN, or some other action.

[0040] In some implementations, the DNS security system may set a timeout that is used to reset the score associated with the FQDN score. The timeout may occur when a request has not occurred for the FQDN within a period, when the FQDN score has been maintained for a period, or at some other interval. When a new request is received, the DNS security service may calculate a new score and update the score based on any subsequent requests.

[0041] FIG. 5 illustrates an operational scenario **500** of updating a trust score associated with a FQDN according to an implementation. Operational scenario **500** includes FQDN notifications **520**, score **540**, maintain score operation **530**, and score timeout operation **532**.

[0042] In deployment, a DNS security service may obtain FQDNs included in DNS requests and determine trust scores associated with each of the FQDNs. The trust scores may be

determined using trust factors related to the reputation of the DNS response, including the services and systems providing the DNS response. The trust factors may include information about the authoritative Name Server, the quantity of requests with a FQDN, changes or information about the A or AAAA records, or some other information.

[0043] Here, the maintain score operation **530** obtains FQDN notifications **520** for a FQDN from a firewall and maintains score **540** for the FQDN. Using the score, the DNS security service may identify traffic rules and provide the traffic rules to a firewall to manage the DNS response and connections from computing systems. In addition to maintaining the score using maintain score operation **530**, the DNS security service may further use score timeout operation **532** to determine when a score should be reset. The timeout may occur when a period expires following the most recent notification with the FQDN, may occur when a period expires following the first notification with the FQDN, or may come after some other interval. Once the timeout occurs, the score may be deleted, and a new score may be generated with the FQDN is identified. Advantageously, if a long period existed between notifications with a FQDN, the factors may change so much that it is preferable that the previous score does not influence the next score.

[0044] FIG. 6 illustrates a DNS security service computing system according to an implementation. Computing system **600** is representative of any computing system or systems with which the various operational architectures, processes, scenarios, and sequences disclosed herein for a firewall may be implemented. Computing system **600** is an example firewall **110**, although other examples may exist. Computing system **600** comprises communication interface **601**, user interface **602**, and processing system **603**. Processing system **603** is linked to communication interface **601** and user interface **602**. Processing system **603** includes processing circuitry **605** and memory device **606** that stores operating software **607**. Computing system **600** may include other well-known components such as a battery and enclosure that are not shown for clarity.

[0045] Communication interface **601** comprises components that communicate over communication links, such as network cards, ports, radio frequency (RF), processing circuitry and software, or some other communication devices. Communication interface **601** may be configured to communicate over metallic, wireless, or optical links. Communication interface **601** may be configured to use Time Division Multiplex (TDM), Internet Protocol (IP), Ethernet, optical networking, wireless protocols, communication signaling, or some other communication format-including combinations thereof. In at least one implementation, communication interface **601** may communicate with a firewall implemented in a computing system or systems. Communication interface **601** may further be used to connect the one or more computing systems to computing elements, such as servers or clients, over the internet.

[0046] User interface **602** comprises components that interact with a user to receive user inputs and to present media and/or information. User interface **602** may include a speaker, microphone, buttons, lights, display screen, touch screen, touch pad, scroll wheel, communication port, or some other user input/output apparatus-including combinations thereof. User interface **602** may be omitted in some examples.

[0047] Processing circuitry **605** comprises microprocessor and other circuitry that retrieves and executes operating software **607** from memory device **606**. Memory device **606** may include volatile and nonvolatile, removable and non-removable media implemented in any method or technology for storage of information, such as computer readable instructions, data structures, program modules, or other data. Memory device **606** may be implemented as a single storage device but may also be implemented across multiple storage devices or sub-systems. Memory device **606** may comprise additional elements, such as a controller to read operating software **607**. Examples of storage media include random access memory, read only memory, magnetic disks, optical disks, and flash memory, as well as any combination or variation thereof, or any other type of storage media. In some implementations, the storage media may be a non-transitory storage media. In some instances, at least a portion of the storage media may be transitory. In no case is the storage media a propagated signal.

[0048] Processing circuitry **605** is typically mounted on a circuit board that may also hold memory device **606** and portions of communication interface **601** and user interface **602**. Operating software **607** comprises computer programs, firmware, or some other form of machine-readable program instructions. Operating software **607** includes request module **608** and trust module **609**, although any number of software modules may provide a similar operation. Operating software **607** may further include an operating system, utilities, drivers, network interfaces, applications, or some other type of software. When executed by processing circuitry **605**, operating software **607** directs processing system **603** to operate computing system **600** as described herein.

[0049] In one implementation, request module **608** directs processing system **603** to obtain a FQDN associated with a domain name system DNS request by a computing device. In some implementations, when a firewall identifies a DNS request with a FQDN, the FQDN is forwarded to the computing system **600**. Once the FQDN is received, trust module **609** directs processing system **603** to generate a first score for the FQDN based on trust factors associated with the FQDN. The trust factors may include, but are not limited to, time delta information between requests for the FQDN, a volume of queries for the FQDN, infrastructure information associated with the A or AAAA records associated with the FQDN, a risk score of the authoritative Name Server, a quantity of record changes associated with the FQDN, SSL certificate comparison with other domains, a quantity of TXT records associated with the FQDN, or some other factor. The trust factors correspond to the infrastructure that is used in responding to the DNS request, wherein the infrastructure may comprise services, servers, certificates, records, and the like associated with responding to the DNS request.

[0050] In some implementations, each of the factors may contribute equally to creating the overall score associated with the FQDN. In other implementations, factors may be weighted differently providing more emphasis on one factor over another. The score may comprise a numerical value (e.g., 0-100), may comprise a letter grade, or may comprise some other scoring mechanism. The information for the factors may be maintained locally, such as information risk status of different Name Servers, or may be stored at least partially on external services or databases that can provide

information about the infrastructure associated with providing the DNS response. The information may then be processed to generate a score.

[0051] Once a score is identified for the FQDN, trust module **609** may direct processing system **603** to determine a traffic rule from a plurality of traffic rules for the DNS request based on the first score. The traffic rule may be based on the score satisfying criteria or a threshold associated with the traffic rule. The traffic rule may indicate that the request is permitted, blocked, should be redirected, or some other action in association with the DNS request. Once the rule is selected, the rule may be provided to the firewall to implement the rule.

[0052] Once an initial score is generated for the FQDN, request module **608** may direct processing system **603** to obtain one or more notifications of one or more additional DNS requests for the FQDN. In response to the one or more notification, trust module **609** directs processing system **603** to update the first score to a second score based on the trust factors associated with the FQDN. As the notifications are received, the trust module **609** may update the information associated with the factors. For example, trust module **609** may update counters associated with the number of DNS requests with the FQDN. Based on the updated factors, a second score can be generated that is an update of the first score. The second score can then be used by trust module **609** to select a traffic rule for the FQDN and provide the selected rule to the firewall for enforcement.

[0053] Although described as separate from the firewall, at least a portion of the operations of the DNS security service may be implemented in the firewall. In other examples, DNS security service **130** may be deployed in a cloud computing center or a server, wherein one or more firewalls may provide FQDNs to the DNS security service for processing and a determination of the traffic rules.

[0054] The included descriptions and figures depict specific implementations to teach those skilled in the art how to make and use the best option. For the purpose of teaching inventive principles, some conventional aspects have been simplified or omitted. Those skilled in the art will appreciate variations from these implementations that fall within the scope of the invention. Those skilled in the art will also appreciate that the features described above can be combined in various ways to form multiple implementations. As a result, the invention is not limited to the specific implementations described above, but only by the claims and their equivalents.

What is claimed is:

1. A method comprising:

receiving a fully qualified domain name (FQDN) identified in a domain name system (DNS) request;
determining trust factors indicative of reliability and reputation of the FQDN;
updating a trust score for the FQDN in accordance with the trust factors, wherein the trust score was generated from previously determined trust factors for the FQDN in response to prior DNS requests received before the DNS request;
after updating the trust score, determining the trust score satisfies an action threshold of a plurality of action thresholds; and
implementing a traffic rule, of a plurality of traffic rules, corresponding to the action threshold, wherein different

traffic rules of the plurality of traffic rules correspond to different action thresholds of the plurality of action thresholds.

2. The method of claim 1, comprising:

receiving the FQDN identified in the prior DNS requests; and

generating the trust score in response to receiving the FQDN identified in the prior DNS requests, wherein the trust score prior to being updated satisfies a second action threshold of the plurality of action thresholds that is different from the action threshold.

3. The method of claim 1, wherein updating the trust score comprises:

increasing the trust score when a factor of the trust factors indicates the trust score should be increased; and
decreasing the trust score when another factor of the trust factors indicates the trust score should be decreased.

4. The method of claim 1, wherein the trust factors comprise at least a time delta between requests for the FQDN and a volume of queries for a period.

5. The method of claim 1, wherein the trust factors comprise popularity information for the FQDN globally.

6. The method of claim 1, wherein the trust factors comprise scores associated with infrastructure to which A or AAAA records for the FQDN point, a risk score of an authoritative Name Server for the FQDN, or a quantity of record changes for the FQDN.

7. The method of claim 1, wherein the plurality of traffic rules comprises a rule to allow traffic, a rule to redirect traffic, and a rule to block traffic.

8. The method of claim 1, comprising:

communicating the traffic rule to a firewall through which one or more computing devices exchange communication traffic.

9. The method of claim 1, comprising:

obtaining information associated with one or more of the trust factors from an external database.

10. An apparatus, comprising:

a storage system;

a processing system operatively coupled to the storage system; and

program instructions stored on the storage system that, when executed by the processing system, direct the apparatus to:

receive a fully qualified domain name (FQDN) identified in a domain name system (DNS) request;

determine trust factors indicative of reliability and reputation of the FQDN;

update a trust score for the FQDN in accordance with the trust factors, wherein the trust score was generated from previously determined trust factors for the FQDN in response to prior DNS requests received before the DNS request;

after updating the trust score, determine the trust score satisfies an action threshold of a plurality of action thresholds; and

implement a traffic rule, of a plurality of traffic rules, corresponding to the action threshold, wherein different traffic rules of the plurality of traffic rules correspond to different action thresholds of the plurality of action thresholds.

11. The apparatus of claim 10, wherein the program instructions direct the processing system to:

receive the FQDN identified in the prior DNS requests;
and

generate the trust score in response to receiving the FQDN identified in the prior DNS requests, wherein the trust score prior to being updated satisfies a second action threshold of the plurality of action thresholds that is different from the action threshold.

12. The apparatus of claim **10**, wherein to update the trust score, the program instructions direct processing system to:

increase the trust score when a factor of the trust factors indicates the trust score should be increased; and

decrease the trust score when another factor of the trust factors indicates the trust score should be decreased.

13. The apparatus of claim **10**, wherein the trust factors comprise at least a time delta between requests for the FQDN and a volume of queries for a period.

14. The apparatus of claim **10**, wherein the trust factors comprise popularity information for the FQDN globally.

15. The apparatus of claim **10**, wherein the trust factors comprise scores associated with infrastructure to which A or AAAA records for the FQDN point, a risk score of an authoritative Name Server for the FQDN, or a quantity of record changes for the FQDN.

16. The apparatus of claim **10**, wherein the plurality of traffic rules comprises a rule to allow traffic, a rule to redirect traffic, and a rule to block traffic.

17. The apparatus of claim **10**, wherein the program instructions direct the processing system to:

communicate the traffic rule to a firewall through which one or more computing devices exchange communication traffic.

18. The apparatus of claim **10**, wherein the program instructions direct the processing system to:

obtain information associated with one or more of the trust factors from an external database.

19. A system comprising:

a firewall computing system configured to:

receive a fully qualified domain name (FQDN) in a domain name system (DNS) request by a computing device;

communicate the FQDN to a DNS security service computing system;

implement a traffic rule received from the DNS security service computing system; and

the DNS security service computing system configured to: in response to receiving the FQDN, determine trust factors indicative of reliability and reputation of the FQDN;

update a trust score for the FQDN in accordance with the trust factors, wherein the trust score was generated from previously determined trust factors for the FQDN in response to prior DNS requests received before the DNS request;

after updating the trust score, determine the trust score satisfies an action threshold of a plurality of action thresholds;

determine a traffic rule, of a plurality of traffic rules, corresponding to the action threshold, wherein different traffic rules of the plurality of traffic rules correspond to different action thresholds of the plurality of action thresholds; and

communicate the traffic rule to the firewall.

20. The system of claim **19**, wherein:

the firewall is configured to:

receive the prior DNS requests; and

communicate the FQDN identified in the prior DNS requests to the DNS security service computing system; and

the DNS security service computing system is configured to generate the trust score the trust score in response to receiving the FQDN identified in the prior DNS requests, wherein the trust score prior to being updated satisfies a second action threshold of the plurality of action thresholds that is different from the action threshold.

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